FLAIR: A Federated Benchmark and Baseline Repository for ICU Prediction Across 12+ Academic Medical Centers Using the Common Longitudinal ICU Format

Manuscript outline — NEJM AI Datasets, Benchmarks, and Protocols

CLIF Consortium
FLAIR Working Group

May 29, 2026

[Working outline. Bulleted content cues sit under every section so drafting prose is paint-by-numbers. Hard envelope: 3000 main-text words, 5 figures+tables combined, 40 references, structured abstract.]

Front matter

- **Title (working):** FLAIR: A Federated Benchmark for ICU Prediction Across 12+ Academic Medical Centers Using the Common Longitudinal ICU Format.
- **Short title** ($\leq$ 50 char): FLAIR: Federated ICU Benchmark.
- **Authors / affiliations:** **TODO:** populate; up to 10 authors; flag equal-contribution and co-senior authorship in a title-page footnote. ORCIDs required.
- **Corresponding author:** **TODO:** name, email, institution.
- **Funding:** **TODO:** grant numbers; restated as a "Funded by" line at the end of the abstract.
- **Conflicts of interest:** **TODO:** ICMJE + Convey forms within 12 months for every author.
- **AI-use disclosure:** **TODO:** name LLM(s); state which sections or artifacts they helped produce. Restate in cover letter.
- **Word / figure / table / reference counts:** Main text *[$\leq$ 3000]*; figures+tables *[$\leq$ 5]*; references *[$\leq$ 40]*.

Structured abstract *[$\approx$ 250 words]*

Background

- ICU prediction models trained on public datasets (MIMIC, eICU, HiRID) routinely fail to generalize beyond the institutions that produced their training data.

- Patient-level data cannot be centralized under HIPAA / IRB constraints, so multi-institutional external validation requires a federated evaluation framework.
- No prior ICU benchmark combines a shared data schema, auditable per-task code, aggregate-only outputs, and TRIPOD-AI reporting.

Methods

- Federated benchmark built atop CLIF v2.1 across N CLIF-consortium sites *[fill N at submission]*.
- Five clinically grounded tasks: T1 ICU daily mortality, T2 ICU daily LTACH discharge, T3 extubation failure <24h, T4 sepsis (antibiotic-anchored) <6h, T5 AKI KDIGO-2 <24h.
- Reference baseline kit: (i) a generic CLIF event tokenizer that maps every clinical event into a token and emits per-window count features, and (ii) two reference models — L2-penalized logistic regression and XGBoost — trained on MIMIC-CLIF and shipped inside the CLIF-WorkBench container for federated evaluation. Sites emit four aggregated JSON reports (`table1`, `tripod_ai`, `fairness`, `curves`).

Results

- Combined cohort: **TODO:** n hospitalizations, n ICU stays, n sites across the consortium. MIMIC-CLIF baseline cohort (development entry point): 84,854 ICU stays at the broadest task (T4 sepsis); 12,438–65,577 stays for the four task-specific cohorts (T1/T2 daily disposition, T3 extubation, T5 AKI); per-task positive-rate range 1.7%–57.4% at decision-row granularity (see Table 1).
- Per-task AUROC / AUPRC (median, IQR across sites) **TODO:** fill from federated runs; calibration drift quantified between MIMIC-CLIF and consortium sites.
- Fairness deltas across sex, race, ethnicity, and age bins; bit- identical SHA-256 hashes across sites confirm protocol fidelity.

Conclusions

- FLAIR enables external validation of ICU AI models without data sharing. Code, container, and per-task schematics released openly. (Funded by **TODO:** ....)

1 Introduction *[≈ 600 words]*

- **P1 — Clinical stakes.** ICU decisions are real-time and high-consequence; rich EHR signal is available but underused.
- **P2 — The saturation problem in public benchmarks.** YAIB (van de Water et al., 2023) and HiRID-ICU-Benchmark (Yèche et al., 2021) harmonize public datasets (MIMIC, eICU, HiRID, AUMCdb); strong internal performance does not imply external generalization.
- **P3 — The privacy barrier.** HIPAA / IRB rules block centralization of PHI; multi-site validation must therefore be federated.

- **P4 — Why prior federated ICU efforts fall short.** Most rely on ad-hoc schemas, lack auditable per-task code, and do not emit TRIPOD-AI-aligned reports.
- **P5 — FLAIR's contribution.** Five clinically grounded tasks; CLIF v2.1 schema-locked code; single-file polars task scripts stamped with SHA-256; **a generic event-tokenizer ETL plus two reference baselines (logistic regression, XGBoost) released as the FLAIR baseline kit**; aggregate-only report bundle; baseline + federated results across the CLIF consortium.
- **Figure 1 callout** *[counts toward 5-item budget]*: *Summary of Contributions* panel modeled on EchoNext-Mini Fig. 1 (Hughes et al., 2026) — one side shows the federated workflow (MIMIC-CLIF → event tokenizer → count features → {logistic regression, XGBoost} → frozen baseline kit (Singularity) → consortium sites → aggregate JSONs); the other shows the five-task taxonomy.

2 Methods *[≈ 1200 words — heaviest section]*

Dataset curation

- CLIF v2.1 data standard maintained by 12+ US academic medical centers; combined coverage ∼2,000 ICU beds nationwide.
- MIMIC-CLIF as public development entry point (∼70k ICU stays; PhysioNet); identical schema guarantees code runs unmodified on consortium data.
- Site enrollment and IRB framework: each site IRB-approved independently; central protocol coordination at the University of Chicago **TODO:** confirm.
- CLIF tables exposed by `flair_benchmark/_clif.py`: `hospitalization`, `adt`, `patient`, `vitals`, `labs`, `respiratory_support`, `medication_admin_intermittent`, `medication_admin_continuous`.
- Data-quality gates inherited from `clifpy` validators; timezone handling per site `clif_config.json`.

Task definitions

Taxonomy. Each task is declared on three axes in its `META` dict: *task class* (outcome / deterioration / intervention), *temporal pattern* (episode / continuous), and *prediction unit* (one-per-stay / one-per-timestep). Full algorithmic specs sit in `docs/task_schematics/` and are included by reference in the Supplement.

- **T1 — ICU Daily In-Hospital Mortality (7 AM).** one-per-timestep, 24 h lookback, daily 07:00 local anchor. Cohort: first ICU stay; discharge ∈ {LTACH, mortality}. *MIMIC-CLIF:* 12,438 stays / 65,989 daily rows; per-row positive rate 57.4%; stays-ever-positive 67.1%.
- **T2 — ICU Daily LTACH Discharge (7 AM).** Complementary label to T1; same cohort and grid. *MIMIC-CLIF:* same 12,438 stays / 65,989 rows; per-row positive rate 42.6%; stays-ever-positive 32.9% (mortality-vs-LTACH disposition is asymmetric).
- **T3 — Extubation Failure within 24 h.** one-per-stay, anchored at the first confirmed extubation event; consensus-window IMV episode detector in `_ventilation_episodes.py`.

Label = death OR reintubation within 24 h post-extubation. *MIMIC-CLIF:* 26,771 extubation episodes; prevalence 5.4% (1,456 / 26,771).

- **T4 — Sepsis (antibiotic-anchored) within 6 h.** one-per-timestep, hourly grid; t_{zero} = first qualifying systemic IV antibiotic; label-leakage gate drops decision points $\geq t_{\text{zero}}$ (Kamran et al., 2024). *MIMIC-CLIF:* 84,854 stays / 7,093,633 hourly rows; per-row positive rate 3.3%; stays-ever-positive 54.9%.
- **T5 — AKI KDIGO-2 within 24 h.** one-per-timestep, hourly grid; SCr-only criterion; Koyner-2018 exclusions (Koyner et al., 2018). *MIMIC-CLIF:* 65,577 stays / 4,135,079 hourly rows; per-row positive rate 1.7%; stays-ever-positive 4.8%.

MIMIC-CLIF cohort snapshot. The numbers below come from `flair extract-metadata -clif-config clif_config_mimic.json` (`flair_benchmark` v2.0.0; HIPAA cell-suppression at $n<10$; SOFA computed over the first 24 h of the first ICU stay via `clifpy`; Charlson and Elixhauser computed from `HospitalDiagnosis` where available — *note: 49–54% of stays are missing comorbidity scores in the MIMIC-CLIF export, a known data-quality gap to flag in the Limitations*).

	T1 mort.	T2 LTACH	T3 ext-fail	T4 sepsis	T5 AKI
ICU stays	12,438	12,438	26,771	84,854	65,577
Decision rows	65,989	65,989	26,771	7,093,633	4,135,079
Positive rate (row)	57.4%	42.6%	5.4%	3.3%	1.7%
Stays ever positive	67.1%	32.9%	—	54.9%	4.8%
Sex: Female / Male	5,546 / 6,892	5,546 / 6,892	10,033 / 16,738	37,552 / 47,302	28,867 / 36,710
Age 65+ share	66.6%	66.6%	54.1%	54.6%	55.1%
SOFA 24h (median, IQR)	5 (2–8)	5 (2–8)	5 (4–8)	3 (1–6)	3 (1–5)
Charlson (median, IQR)	4 (2–5)	4 (2–5)	2 (0–4)	2 (0–4)	2 (0–4)

Table 1: MIMIC-CLIF cohort snapshot per task, generated by `flair extract-metadata`. *Decision rows* = one-per-stay for T3 (extubation episode), one-per-timestep otherwise (daily at 07:00 for T1/T2; hourly for T4/T5). *Age 65+* share is over distinct ICU stays. *Stays ever positive* is omitted for T3 because the task is one-per-stay (the row-level rate is the stay-level rate).

[This table is a drafting hook for Results—Table 1 and Methods—Cohort. It is not part of the 5-item NEJM AI figure/table budget; the submission Table 1 will mirror this layout across consortium sites with site-level median/IQR.]

Splits. Temporal date-based train/test split; MIMIC fallback 70/30 when split dates not specified.

Cohort parquet schema. Ten columns: `hospitalization_id`, `window_start`, `window_end`, `<label>`, `split`, `age_at_admission`, `sex_category`, `race_category`, `ethnicity_category`, and the task-specific `label_*` column.

Federated protocol

- Two-layer privacy model: (i) CLIF-WorkBench Singularity container — no internet, no root, immutable read-only image; (ii) aggregate-only outputs from `flair build-report`.

- Library-enforced HIPAA safeguards: cell suppression in `table1.json` for counts <10; subgroup drop in `fairness.json` for subgroups with $n < 10$; calibration-bin suppression in `curves.json`.
- Audit chain: SHA-256 of each task script is stamped into every output JSON; the central aggregator verifies byte-identical task code across sites.
- Submission contract: user appends `y_pred`/`y_prob` to the cohort parquet; `flair build-report` emits exactly four JSON files. No patient-level data ever leaves the site. For the FLAIR baseline kit, the container additionally ships the tokenizer and frozen weights; sites invoke `flair predict-baseline <task>` instead of writing their own feature/model code.

Baseline kit

Generic event tokenizer. Each row in a CLIF event table (`vitals`, `labs`, `medication_admin_intermittent`, `medication_admin_continuous`, `respiratory_support`) is mapped to a deterministic token of the form `<table>.<category>` (e.g., `vitals.heart_rate`, `labs.creatinine`, `meds.norepinephrine`). For each cohort window, FLAIR emits a sparse token → integer-count vector counting events in [`window_start`, `window_end`]. The tokenizer is config-free and exhaustive — every category present in the input flows through with no per-task curation. Because tokens are derived from CLIF schema-locked category fields, the same code runs unchanged on MIMIC-CLIF and consortium sites. Module: `flair_benchmark/baseline/tokenizer.py`.

Logistic-regression baseline. Sparse count features → L2-penalized logistic regression (`sklearn`); submitted as `y_prob` for the binary tasks.

XGBoost baseline. The same sparse count features → XGBoost classifier with default depth and learning rate; submitted as `y_prob`.

Both models are trained on MIMIC-CLIF (70/30 temporal split via `_split.py`) for each of T1–T5, serialized into the CLIF-WorkBench Singularity container, and shipped to each consortium site. Hyperparameters, the token dictionary, and the missingness convention (absent token = zero count) are deferred to the Supplement (Model Cards + Datasheet).

Reporting and statistical analysis

- TRIPOD+AI (Collins et al., 2024) compliance via `flair_benchmark/report/tripod_ai.py` auto-population.
- Primary metrics: AUROC, AUPRC; 95% CIs via 1000-iteration bootstrap.
- Calibration: slope, intercept, Brier score, reliability curves (downsampled raw points in `curves.json`).
- Fairness: per-subgroup AUROC and parity gaps across sex, race, ethnicity, and age bins.
- Central dashboard renders ROC / PR / calibration curves from the aggregated `curves.json` without re-accessing patient data.

3 Results *[≈ 700 words]*

Cohort

- **Table 1.** Combined cohort characteristics across N sites (HIPAA-suppressed cells where applicable), stratified by split and label.
- **Figure 2 callout:** CONSORT-style flow diagram per task (raw → cohort filters → train/test). Mirrors EchoNext-Mini Fig. 2 layout.
- Reported counts: total hospitalizations, total ICU stays per task, per-task prevalence, demographic distribution.

Baseline performance on MIMIC-CLIF

- Per-task internal test AUROC / AUPRC with 95% bootstrap CIs, reported separately for each model family (logistic regression, XGBoost).
- Calibration slope / intercept / Brier score per task per model family.

Federated performance across the CLIF consortium

- **Table 2.** Per-task site-level summary (median, IQR, range for AUROC / AUPRC across N sites), with a `model_family` column so each task carries one row per family per site.
- **Figure 3.** Forest plot of per-site AUROC by task with two glyphs per task (logistic regression vs XGBoost); vertical reference line at the MIMIC-CLIF internal benchmark for that family.
- Quantify calibration drift between MIMIC-CLIF and held-out consortium sites, separately by model family.

Fairness

- **Figure 4.** Heatmap of per-subgroup AUROC (rows = tasks, columns = subgroups) with parity-gap shading. Cells with $n < 10$ are suppressed.

Reproducibility evidence

- Bit-identical SHA-256 task-script hashes recorded in every output JSON across sites; the central aggregator's audit log confirms protocol fidelity.

4 Discussion *[≈ 400 words]*

- **Generalization.** How much do MIMIC-trained baselines drop when evaluated on held-out consortium sites? Quantify and contextualize.

- **Positioning vs. prior benchmarks.** YAIB (van de Water et al., 2023) and HiRID-ICU-Benchmark (Yèche et al., 2021) harmonize public sources only; FLAIR's differentiator is federation across private data with an audit chain.
- **Limitations.** `clifpy` data-quality dependency; T5 CKD-stage exclusion not yet implemented (CLIF `Diagnoses` table not exposed in FLAIR loaders); T4 antibiotic drug-list curation is an open dependency; baseline tokens are pure counts within the prediction window with no time-of-occurrence, magnitude, or trend features (deliberate floor; richer encodings are left to user submissions); no urine-output criterion for AKI.
- **Bias and fairness.** Consortium sites self-select; rural and safety-net hospitals are under-represented.
- **Intended use.** External validation tool for development and method comparison — *not* a leaderboard endorsing production deployment.
- **Regulatory framing.** Per FDA CDS guidance and IMDRF SaMD clinical-evaluation guidance (see `docs/paper/lit/`), the benchmark targets the "inform clinical management" use class, not autonomous decision-making.

5 Conclusions *[≈ 120 words]*

- One paragraph. FLAIR delivers a federated, schema-locked, audit-traceable ICU benchmark **paired with a reusable baseline kit (event tokenizer + logistic regression + XGBoost) that defines the floor of expected performance at every site**; release includes code, container, baseline weights, and per-task LaTeX schematics. Community extension via GitHub PRs is invited.

Required statements

- **Data and Code Availability.** GitHub `clif-consortium/FLAIR` (Apache 2.0); MIMIC-CLIF on PhysioNet; baseline weights on Zenodo **TODO:** DOI placeholder; Singularity image tag on GHCR **TODO:** tag placeholder.
- **IRB.** Each consortium site obtained independent IRB approval; central protocol coordination at the University of Chicago **TODO:** confirm.
- **Funding.** **TODO:** grants.
- **Author Contributions.** CRediT taxonomy slots **TODO:** fill.
- **Conflicts of Interest.** **TODO:** disclosures.
- **AI-Use Disclosure.** **TODO:** which LLM helped where.

Supplementary Appendix *[no NEJM AI hard cap]*

- Per-task LaTeX schematics (T1–T5), compiled from `docs/task_schematics/`; the sepsis-T4 schematic (Kamran et al., 2024) already exists and is the template.

- Datasheet for Datasets (Gebru-style).
- Model Cards (Mitchell-style) for the two baseline models (logistic regression and XGBoost).
- Tokenizer specification — deterministic `<table>.<category>` mapping; vocabulary size on MIMIC-CLIF; missingness convention (absent token = zero count); per-token frequency histograms.
- Full TRIPOD+AI checklist.
- Hyperparameter table, feature dictionary, missingness table.
- Per-site Table-1 details (cells with $n \geq 10$).

References

Gary S. Collins, Karel G. M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B. Reitsma, Maarten van Smeden, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.

John Weston Hughes, Linyuan Jing, Joshua Finer, Dustin Hartzel, Christopher Kelsey, Aaron Long, Daniel Rocha, Jeffrey Ruhl, Timothy Poterucha, and Pierre Elias. EchoNext-Mini: A dataset and baseline AI model for detecting structural heart disease from electrocardiograms. *NEJM AI*, 3(5), 2026. doi: 10.1056/AIdbp2500516.

Fahad Kamran, Donna Tjandra, Andrew Heiler, Jessica Virzi, Karandeep Singh, Jessie E. King, Thomas S. Valley, and Jenna Wiens. Evaluation of sepsis prediction models before onset of treatment. *NEJM AI*, 1(3), 2024. doi: 10.1056/AIoa2300032.

Jay L. Koyner, Kyle A. Carey, Dana P. Edelson, and Matthew M. Churpek. The development of a machine learning inpatient acute kidney injury prediction model. *Critical Care Medicine*, 46(7): 1070–1077, 2018. doi: 10.1097/CCM.0000000000003123.

Robin van de Water, Hendrik Nils Aurel Schmidt, Paul Elbers, Patrick Thoral, Bert Arnrich, and Patrick Rockenschaub. Yet another ICU benchmark: A flexible multi-center framework for clinical ML. *arXiv preprint arXiv:2306.05109*, 2023. doi: 10.48550/arXiv.2306.05109.

Hugo Yèche, Rita Kuznetsova, Marc Zimmermann, Matthias Hüser, Xinrui Lyu, Martin Faltys, and Gunnar Rätsch. HiRID-ICU-Benchmark — a comprehensive machine learning benchmark on high-resolution ICU data. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.